Question 2

A1 Procedure OK (i.e. find m_B and acc^n of A or B; change m_B and repeat). **1**
An experiment must have been described for this mark to be awarded.
This mark can be scored even if the method is unworkable.

A2 Diagram of workable arrangement to find acceleration **1**
(e.g. object falls between two markers/light gates/smart pulley at top)
If the diagram is not very detailed refer to text.

A3 Measurement of mass m_B (e.g. using balance/Newton meter/calibrations on masses) **1**

B1 Valid method of measuring time **1**
Accept stopwatch; ticker-tape; light gates; motion sensors and dataloggers; smart pulley etc..
Unworkable methods will not score this mark.

B2 Correct measurements taken to find acceleration **1**
(e.g. measure a distance and $u = 0$ (if distance/time method used)
spacing of successive dots on ticker-tape
some detail of sampling rate if motion sensor/datalogger used)

B3 Use of results to calculate acceleration **1**
(e.g. substitute into $s = ut + \frac{1}{2}at^2$; $a = 25(x_2 - x_1)$ etc..)
If motion sensor used then acceleration obtained from monitor.

C1 Any one safety precaution **1**
(e.g. Catch falling mass in bucket of sand
Care needed to prevent mass B from coming over the top of the pulley
Whiplash from breaking wires etc.
Clamp retort stand to prevent it from falling over.
Do not allow vague 'safety goggles'. Insist on a reason being given.

D1/2/3 Any further good design features **3**
Some of these might be:
Method of supporting the pulley
Mention of friction in the pulley/oil pulley/smooth pulley
Use large distance (to reduce percentage uncertainty)
Limitations of stopwatch methods
Vary s and measure t; use graph to find a
Repeat the experiment to find values of a for each value of m_B
Some detail about the timing circuit (e.g. stop terminals on timer connected to double pole switch and electromagnet).

10 marks in total.

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